Magic Attention

A Composable Framework for Exploring Warp-Specialized Attention on Blackwell

Speaker: Manish Gupta

Magic

Contributors & Acknowledgements

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Technical References	FlashAttention, FA1-4, CUTLASS example 77
Infrastructure & Tools	CUTLASS/CuTe, NVCC, PTXAS, NCU, CompileIQ

1 Magic 2 NVIDIA

Magic-Attention

- **Develop** a composable framework of components for attention kernels on NVIDIA Blackwell
- **Analyze** trade-offs with different warp-specialization schedules, Tensor Memory partitioning, and instruction mix
- **Cover** functionality & performance, over a wide range of training, inference, and RL workloads scenarios

Attention on Blackwell

Level	Concept	Variations
Attention Algorithm-specific	SeqlenKind	Fixlen • Varlen • VarlenPaged
	MaskKind	Causal • NoMask • Sliding Window • Strided
	GQAPacking	Packed • Not Packed
	Density	Dense • Vector Sparse • Block Sparse • Hierarchical
Scheduling-specific	Tile Scheduler (Grid-level scheduling)	Static Persistent • Dynamic Persistent
	Kernel Schedule (Warp-specialization)	1xSoftmax • 2xSoftmax • Exp-Transform • 2xQ
Low-level hardware-specific	Store output	TMA-STORE • STG
	Math	UMMA.2CTA • UMMA • MMA.SYNC • Mixed
	Load input	TMA-LOAD • ASYNC-COPY • Mixed
	Dtype	BF16 • F8 • F4 • Mixed
	Tensor Memory (TMEM) partitioning	128x512x32b TMEM with different partitioning

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	Dtype	
	Tensor Memory (TMEM) partitioning	

136 forward attention kernel variants for one dtype

16 forward attention foundational kernel variants in this talk

Attention on Blackwell

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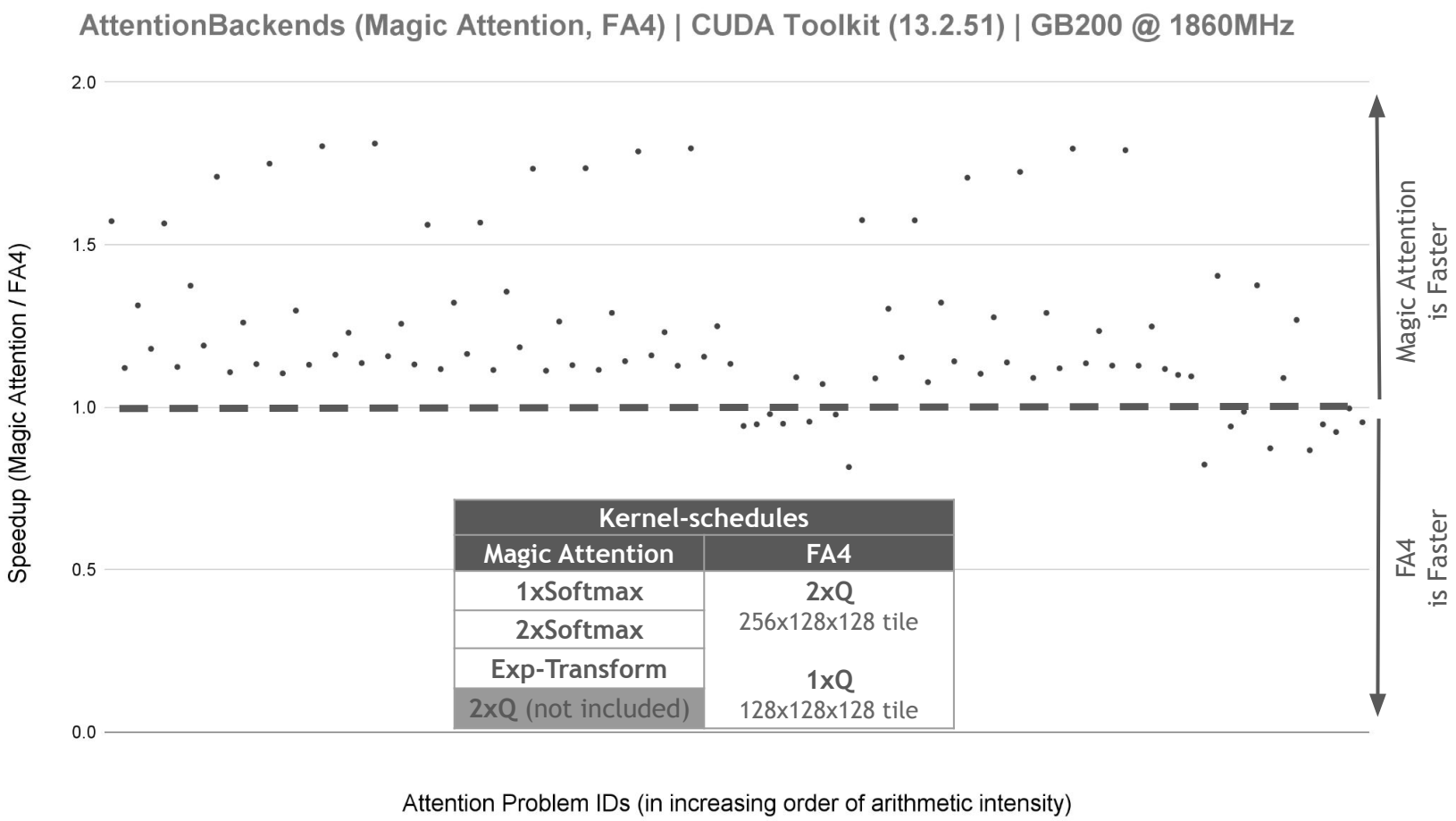
	Dtype	
	Tensor Memory (TMEM) partitioning	

CUDA Graph
out of the box
for all variants

Static Persistent
SM carve-out support

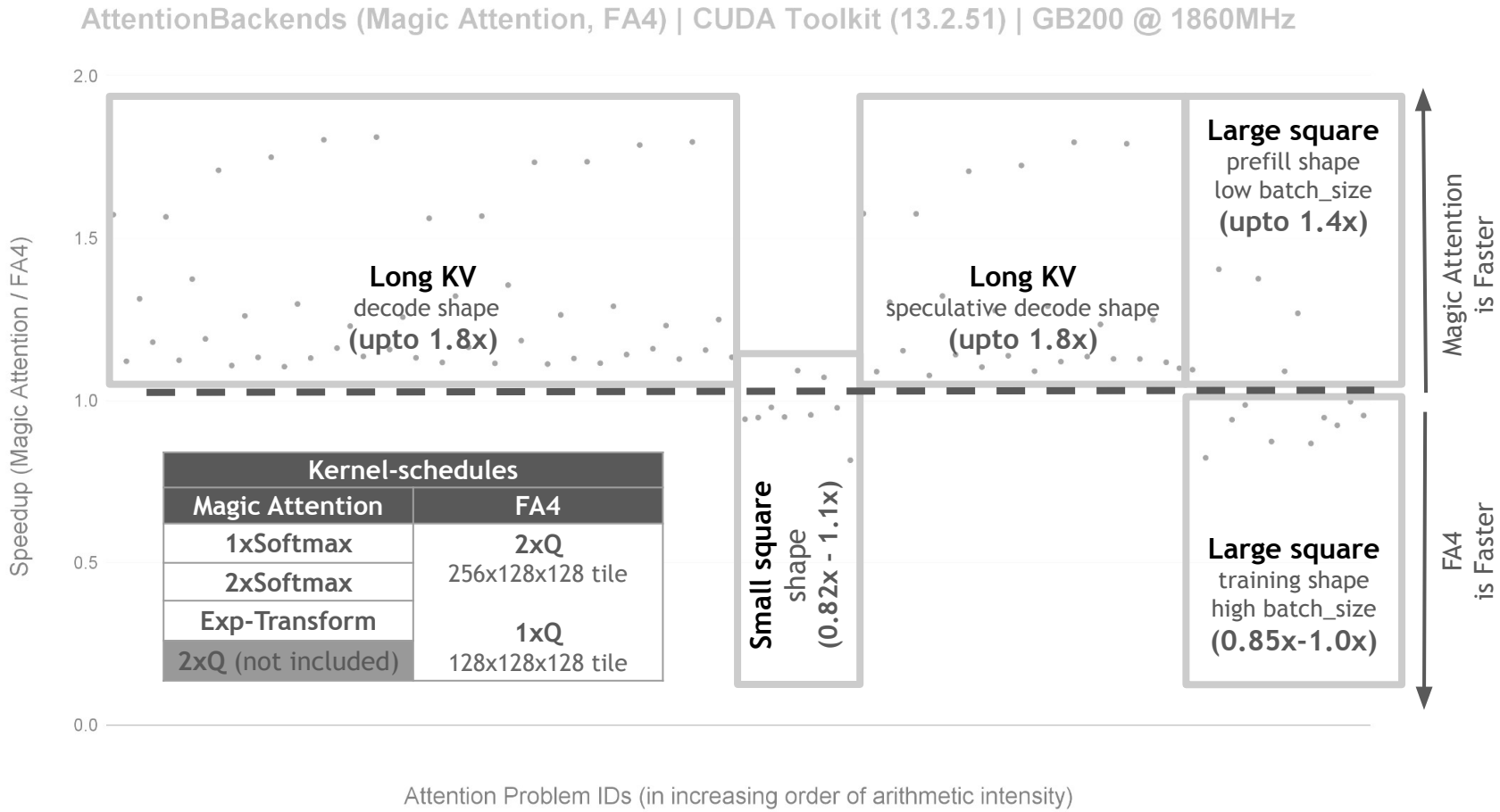
Dynamic Persistent
Cluster Launch Control (CLC)
Efficient draining of oversized grid

Performance on GB200



Magic

Performance on GB200



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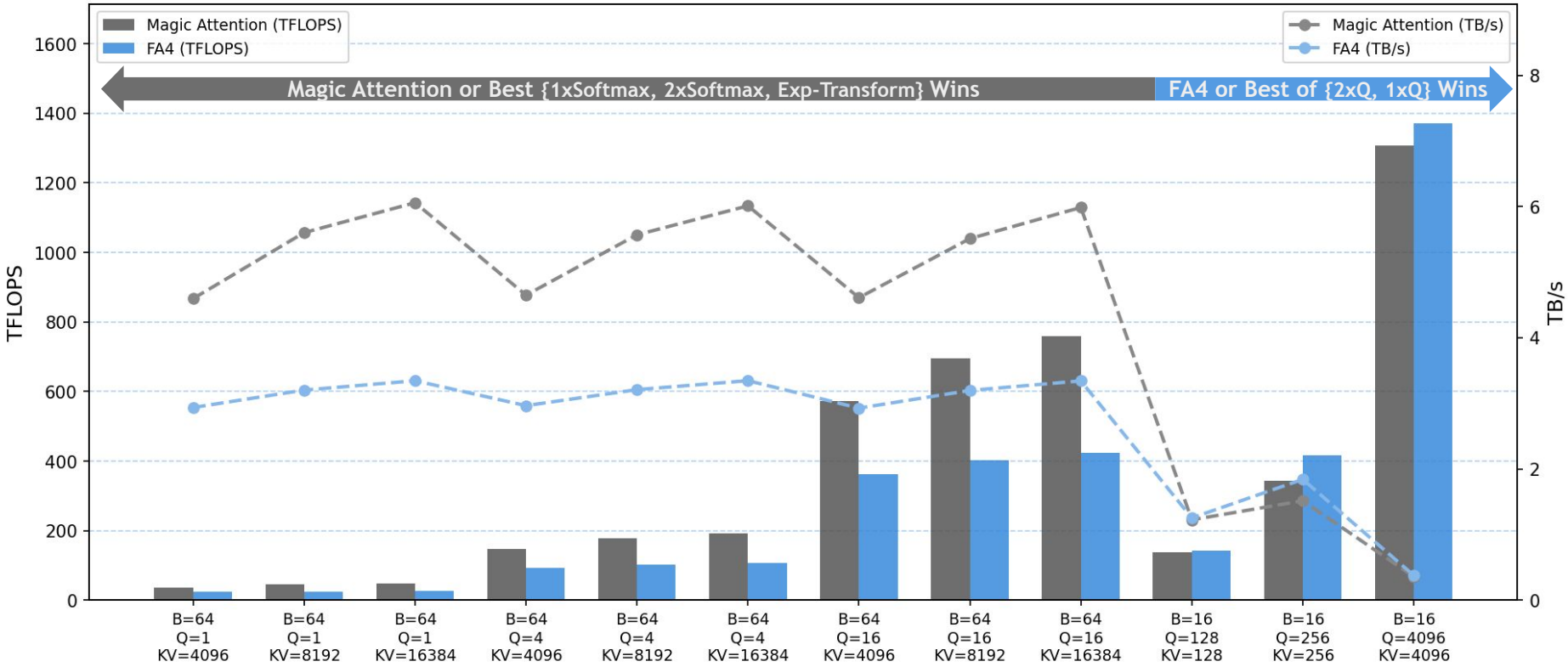
Performance on GB200

AttentionBackends=(Magic Attention, FA4) | CTK=13.2.51 | CLK=1860MHz | warmup-time=25ms | profiling-time=1000ms | cudagraph enabled



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Agenda

- Attention Algorithm
- Tiling Attention on a GPU
- Convolution-inspired Runtime GQA Packing
- Attention on NVIDIA Blackwell Architecture
- Feeding the Datapath on NVIDIA Blackwell GB200

Attention Operation

$$\mathbf{QK} = \mathbf{Q} \mathbin{@} \mathbf{K}^{T}$$

$$\mathbf{E} = \mathrm{Softmax}(\mathbf{QK}) \,/\, \mathrm{sqrt}(H)$$

$$\mathbf{A} = \mathrm{Transform}(\mathbf{E})$$

$$\mathbf{AV} \mathrel{+}= \mathbf{A} \mathbin{@} \mathbf{V}$$

Attention Operation

$$QK = Q \mathbin{@} K^T$$

$$E = \mathrm{Softmax}(QK) / \mathrm{sqrt}(H)$$

$$\boxed{A = \mathrm{Transform}(E)}$$

$$AV \mathrel{+}= A \mathbin{@} V$$

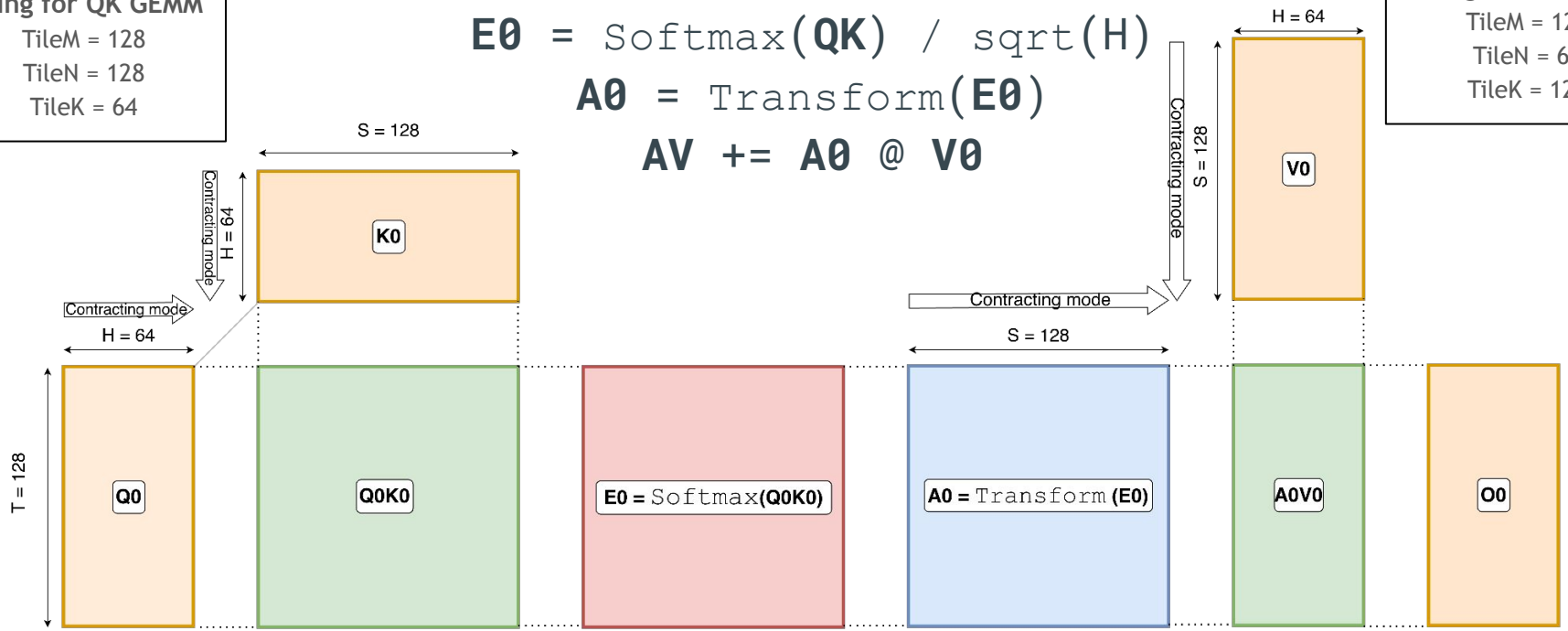
- `Transform` does not alter the core attention algorithm
- **Cleaner abstractions** enables **crazier warp-specializations**

Attention Tiling - Journey of a Tile

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

$$QK = Q0 \ @ \ K0^T$$
$$E0 = \mathrm{Softmax}(QK) \ / \ \mathrm{sqrt}(H)$$
$$A0 = \mathrm{Transform}(E0)$$
$$AV \mathrel{+}= A0 \ @ \ V0$$

Tiling for QK GEMM
TileM = 128
TileN = 64
TileK = 128



T : Query tokens (128)
S : Key-value tokens (128)
H : Head dimension (64)

N : Number of Q heads (1)
K : Number of KV heads (1)
G : Number of Q heads per KV heads (N // K = 1)

Notations from the JAX ebook [How To Scale Your Model](#)'s Transform Accounting section

Magic

Attention Tiling - Journey of a Tile

$$QK = Q0 \ @ \ K0^T$$

$$QK = Mask(QK)$$

$$E0 = Softmax(QK) \ / \ sqrt(H)$$

$$A0 = Transform(E0)$$

$$AV = Correction(AV)$$

$$AV \mathrel{+}= A0 \ @ \ V0$$

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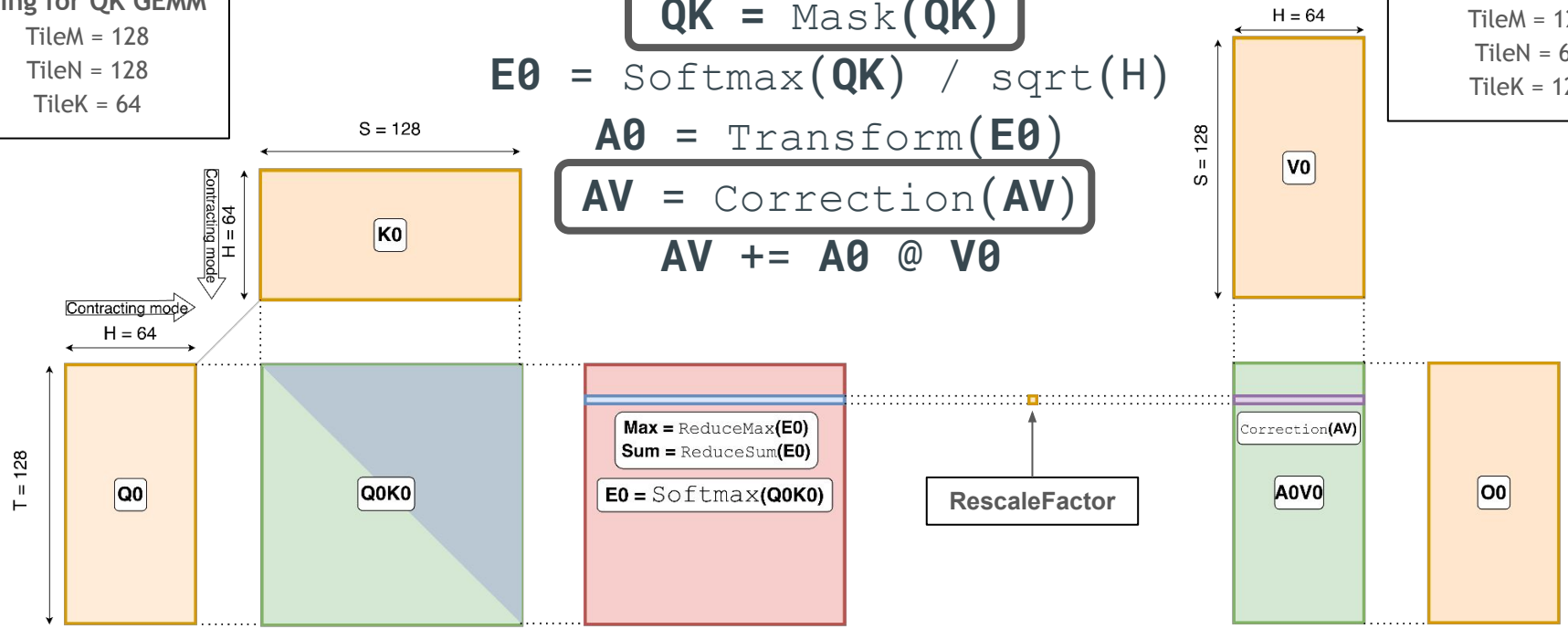

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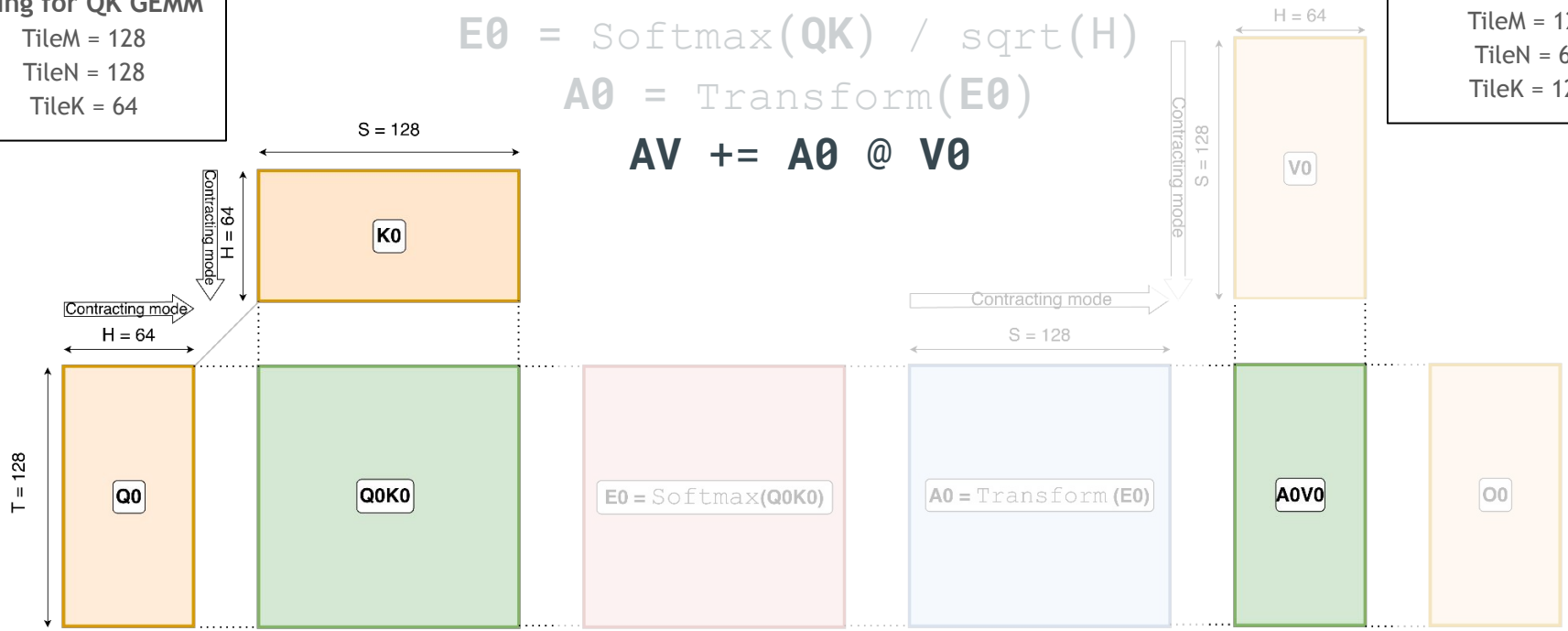
FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness, Tri Dao et. al.

Attention Tiling - Journey of a Tile

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

$$QK = Q0 \ @ \ K0^T$$
$$E0 = Softmax(QK) \ / \ sqrt(H)$$
$$A0 = Transform(E0)$$
$$AV \ += \ A0 \ @ \ V0$$

Tiling for QK GEMM
TileM = 128
TileN = 64
TileK = 128



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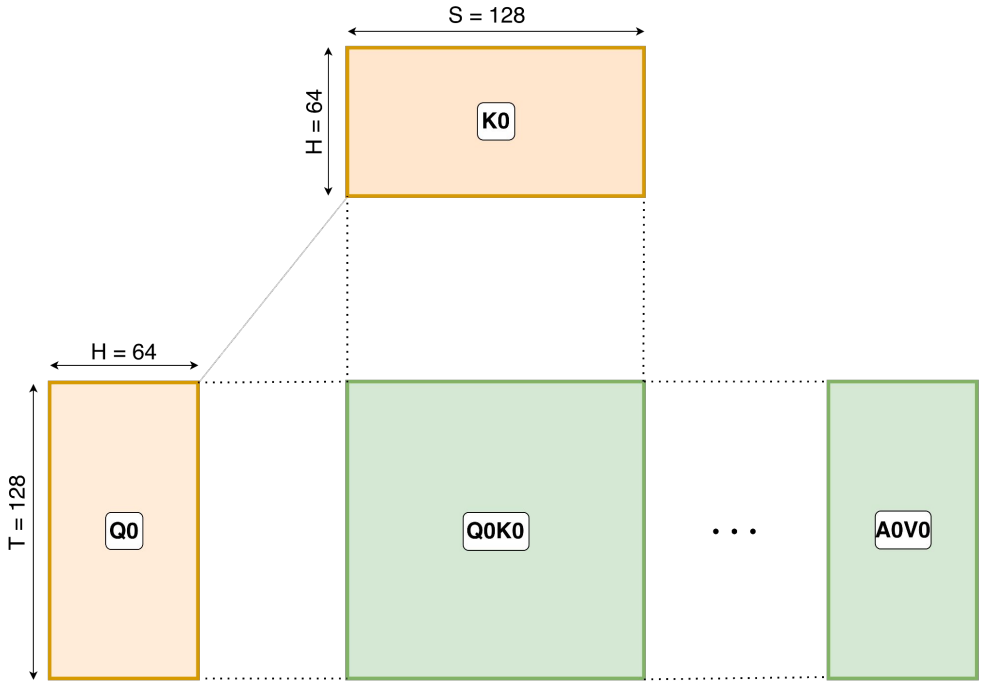
Attention Tiling - 1 Q Tile, 1 KV Tile

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

Tiling for QK GEMM
TileM = 128
TileN = 64
TileK = 128



T : Query tokens (128)
S : Key-value tokens (128)
H : Head dimension (64)

N : Number of Q heads (1)
K : Number of KV heads (1)
G : Number of Q heads per KV heads (N // K = 1)

Attention Tiling - 2 Q Tile, 4 KV Tile

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

Tiling for QK GEMM
TileM = 128
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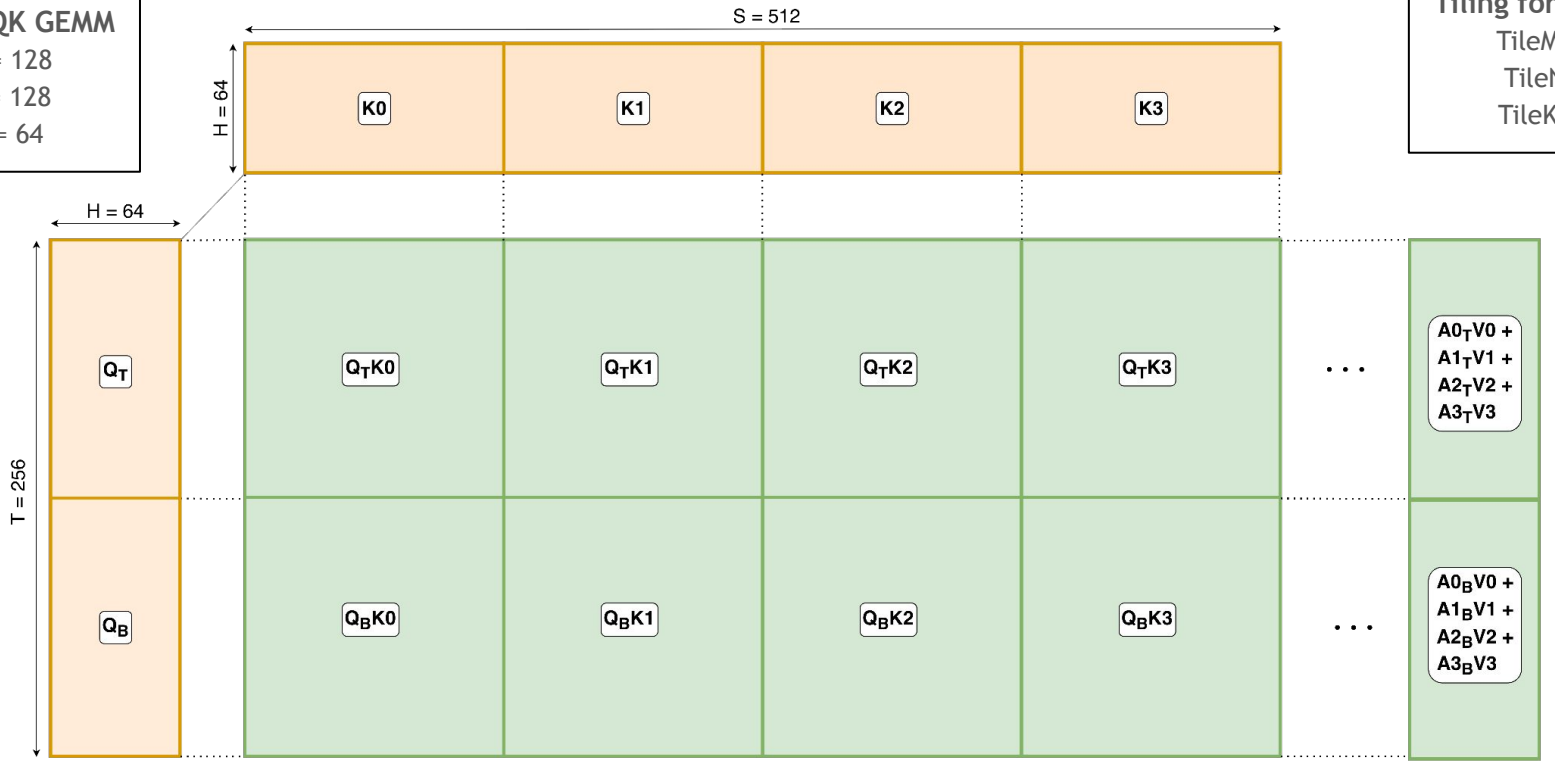
TileK = 128



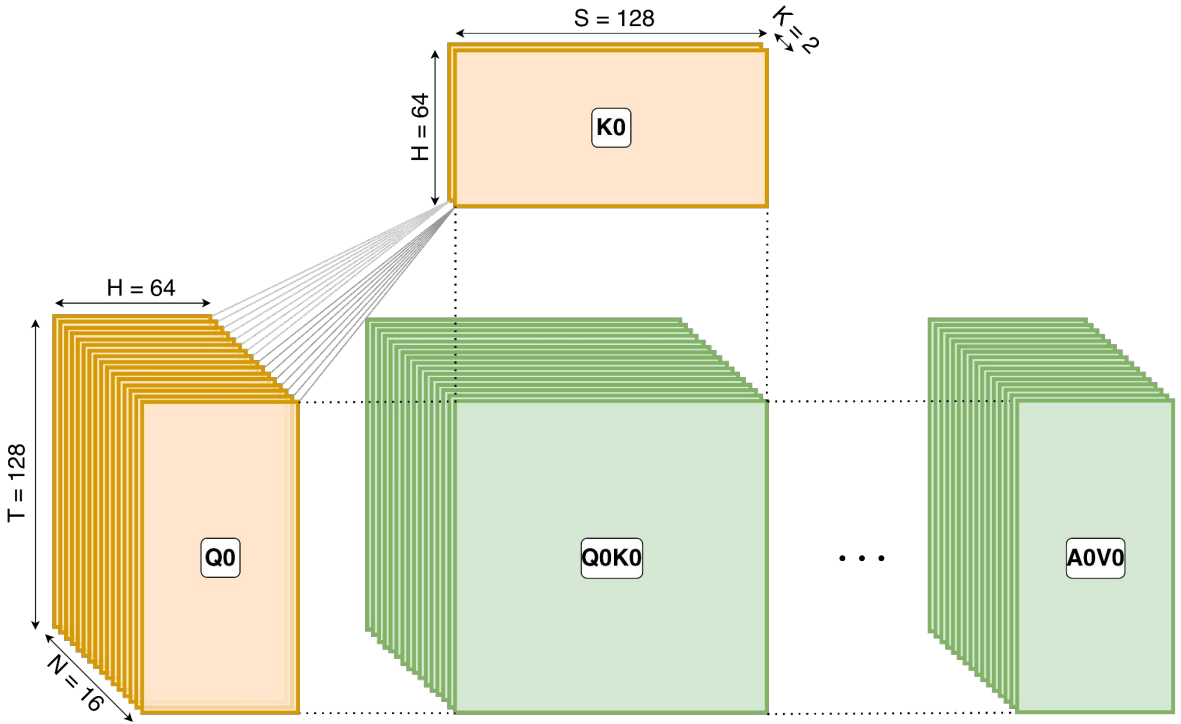
T : Query tokens (256)
S : Key-value tokens (512)
H : Head dimension (64)

N : Number of Q heads (1)
K : Number of KV heads (1)
G : Number of Q heads per KV heads (N // K = 1)

Attention Tiling - Multi-dimensional Tensors

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

Tiling for QK GEMM
TileM = 128
TileN = 64
TileK = 128



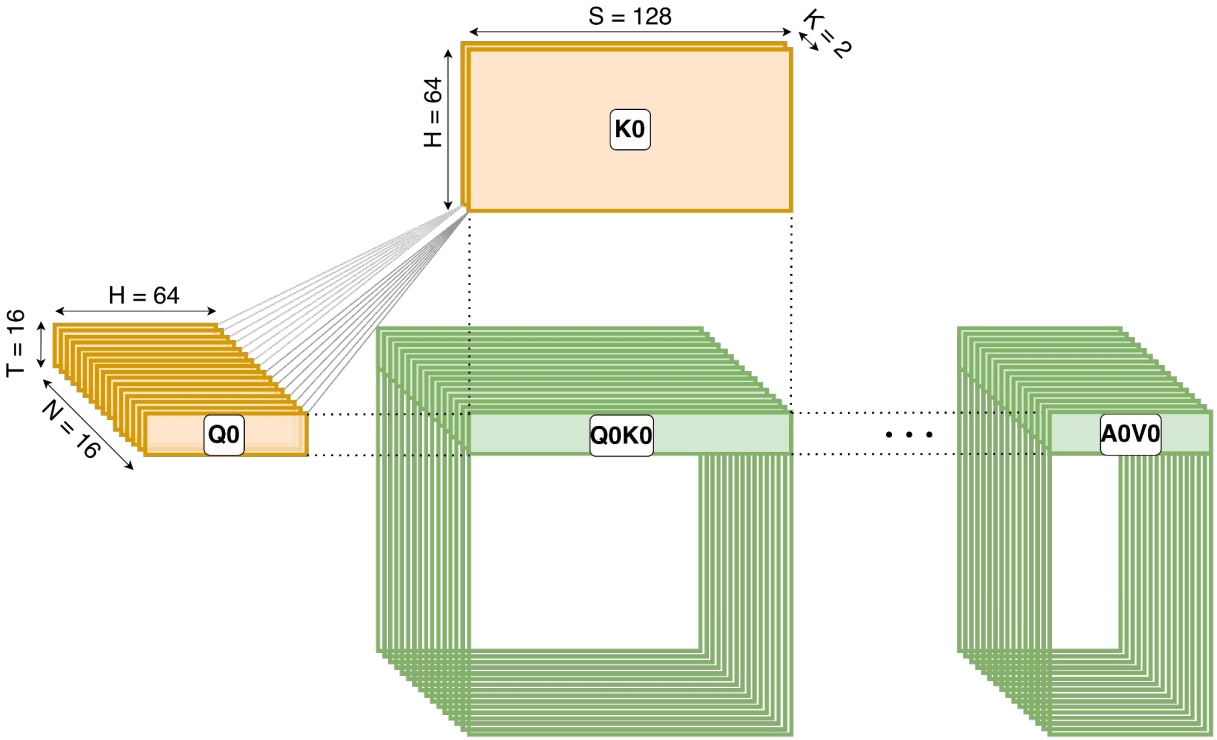
T : Query tokens (128)
S : Key-value tokens (128)
H : Head dimension (64)

N : Number of Q heads (16)
K : Number of KV heads (2)
G : Number of Q heads per KV-heads (N // K = 8)

Attention Tiling - Multi-dimensional Tensors

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

Tiling for QK GEMM
TileM = 128
TileN = 64
TileK = 128



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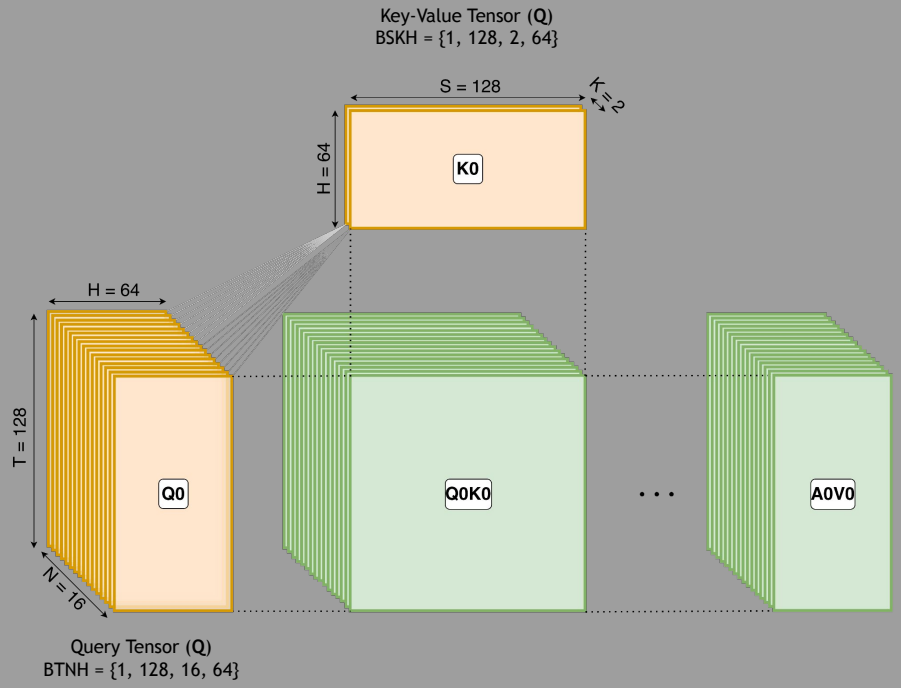
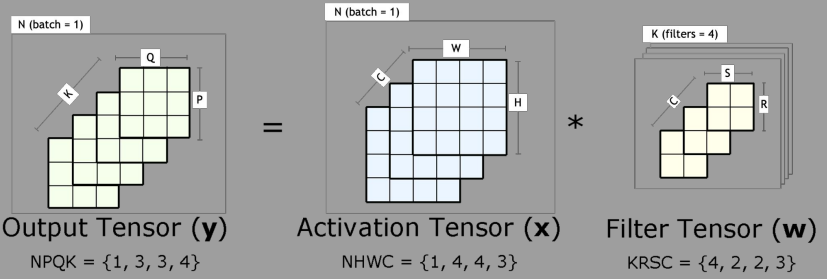
G : Number of Q heads per KV-heads (N // K = 8)

Digression from Attention to Convolution



B (Batch) = 1

Attention



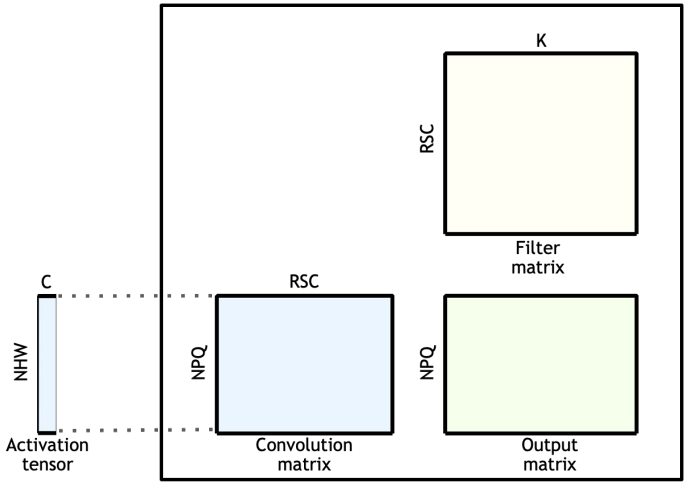
N (Batch) = 1

Convolution

4D Tensors to 2D Matrices at **Runtime**

The convolution operation on 4D tensors can be mapped as matrix-multiply operation on 2D matrices

4D Tensors	**2D Matrix**
y = CONV(**x**, **w**)	**C** = **A** @ **B**
x[N,H,W,C] : 4D activation tensor	**A**[NHW, RSC] : 2D convolution matrix
w[K,R,S,C] : 4D filter tensor	**B**[RSC, K] : 2D filter matrix
y[N,P,Q,K] : 4D output tensor	**C**[NPQ, K] : 2D output matrix

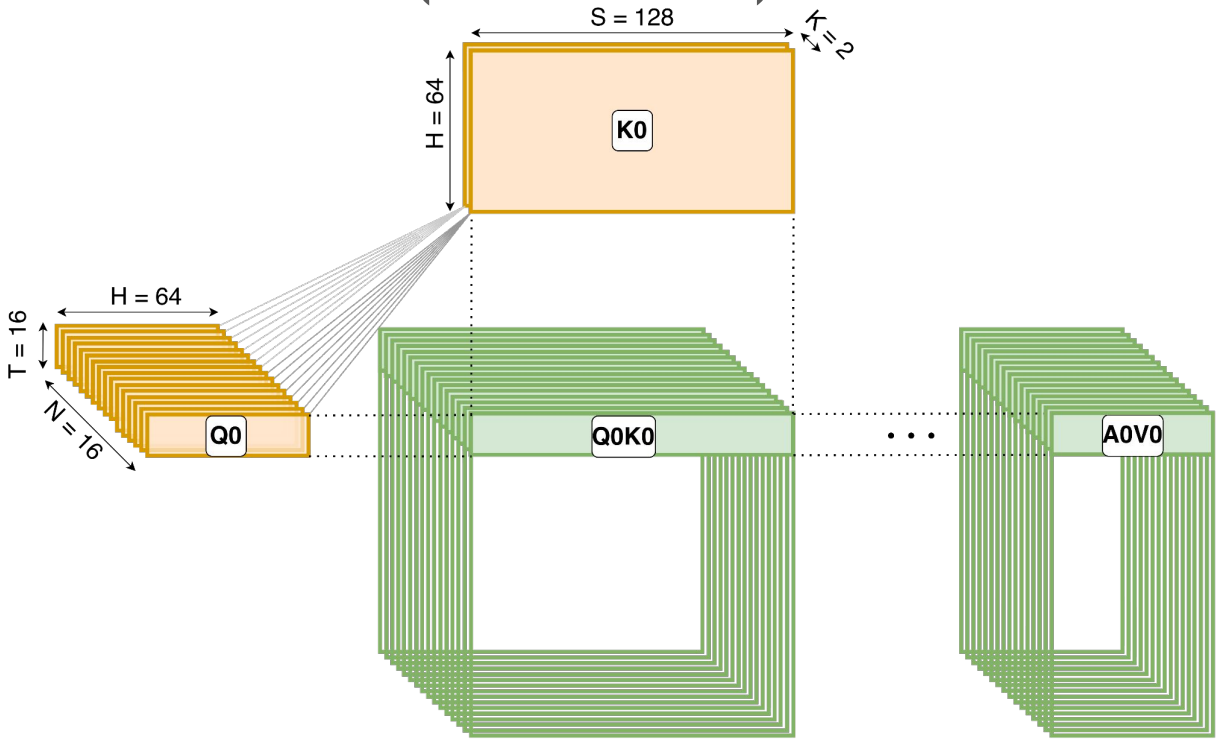


Source: Accelerating Convolution with Tensor Cores in CUTLASS, GTC 2021

Attention Tiling - Multi-dimensional Tensors (Not Packed)

Tiling for QK GEMM
TileM = 128
TileN = 128
TileK = 64

Tiling for QK GEMM
TileM = 128
TileN = 64
TileK = 128



T : Query tokens (16)
S : Key-value tokens (128)
H : Head dimension (64)

N : Number of Q heads (16)
K : Number of KV heads (2)
G : Number of Q heads per KV-heads (N // K = 8)

Attention Tiling - Multi-dimensional Tensors (Packed)

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TileK = 64

Tiling for QK GEMM
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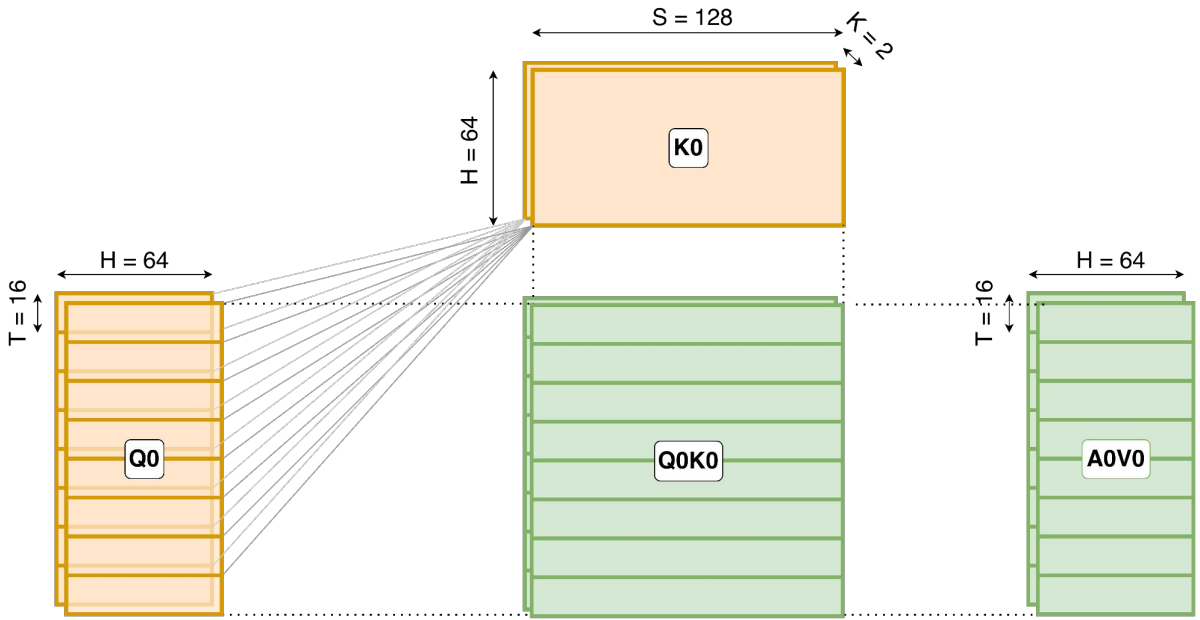
TileK = 128



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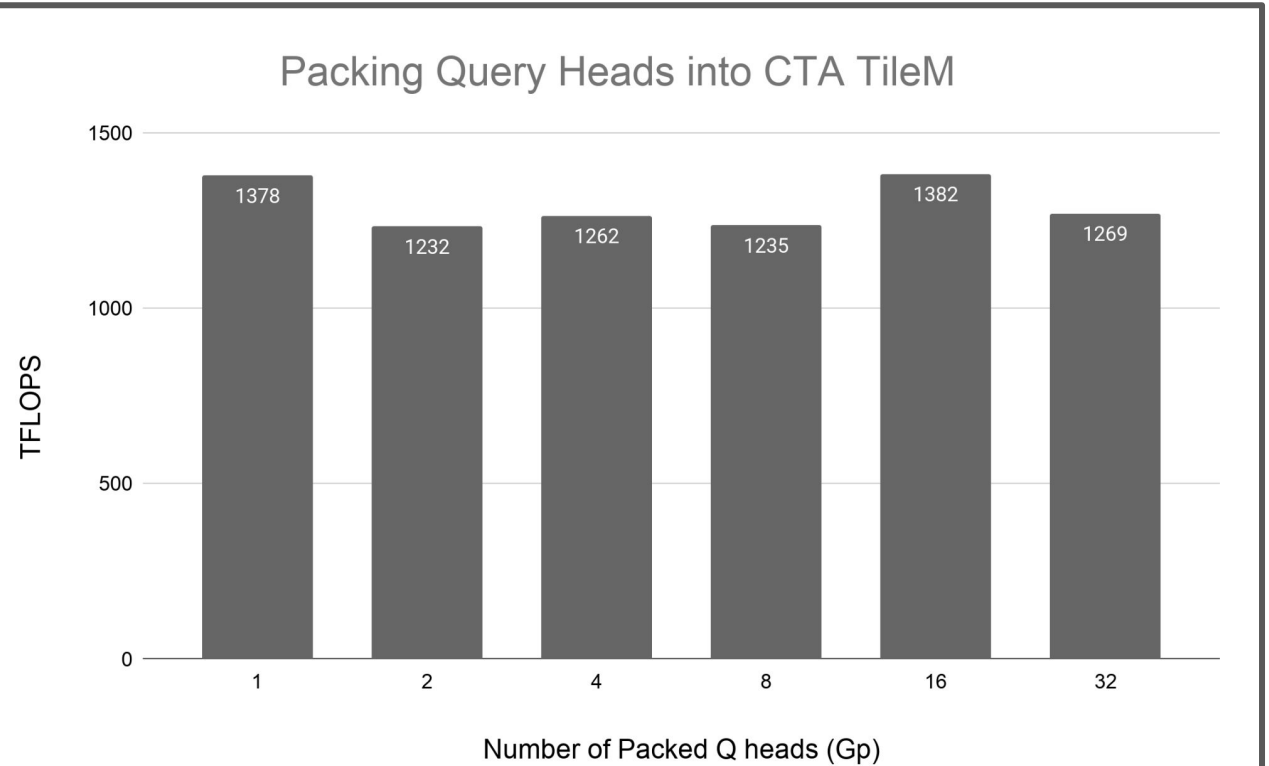
S : Key-value tokens (128)
H : Head dimension (64)

N : Number of Q heads (16)
K : Number of KV heads (2)
G : Number of Q heads per KV-heads (N // K = 8)

Gp : Number of Packed Q heads (8)
Gu : Number of Un-packed Q heads (1)
G = Gp * Gu (8)

Performance with Runtime GQAPacking



Takeaways

1. Peak performance isn't always at max packing.
2. **GQAPacking *improves* CTA utilization** but ***degrades*** **grid parallelism.**
3. **Split-KV** with GQAPacking ***restores*** **grid parallelism.**
4. **Magic-Attention** introduces a convolution-inspired **runtime knob to tune GQAPacking for peak performance.**

T : Query tokens (48)
S : Key-value tokens (8192)
H : Head dimension (128)

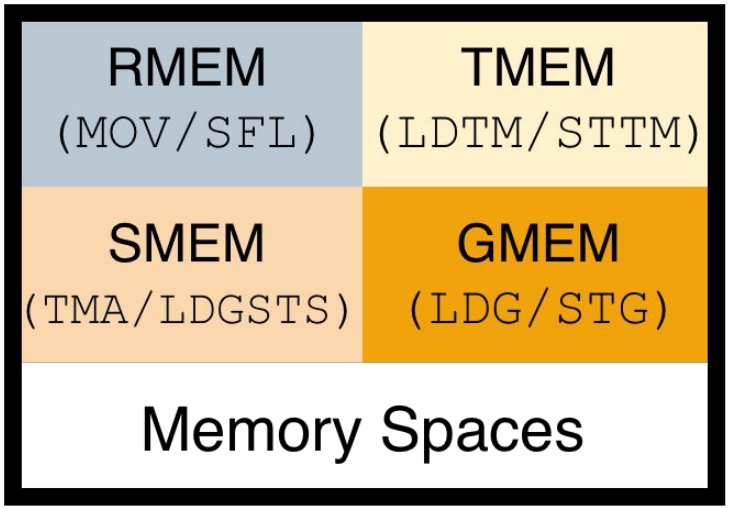
N : Number of Q heads (32)
K : Number of KV heads (1)
G : Number of Q heads per KV-heads (N // K = 32)

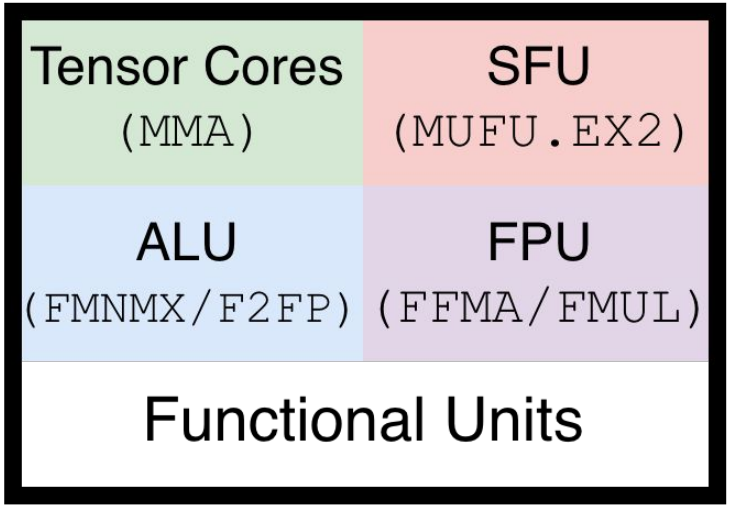
Gp : Number of Packed Q heads (1,2,4,8,16,32)
Gu : Number of Unpacked Q heads (32,16,8,4,2,1)
G = Gp * Gu

Agenda

- Attention Algorithm
- Tiling Attention on a GPU
- Convolution-inspired Runtime GQA Packing
- Attention on NVIDIA Blackwell Architecture
- Feeding the Datapath on NVIDIA Blackwell GB200

NVIDIA Blackwell GPU



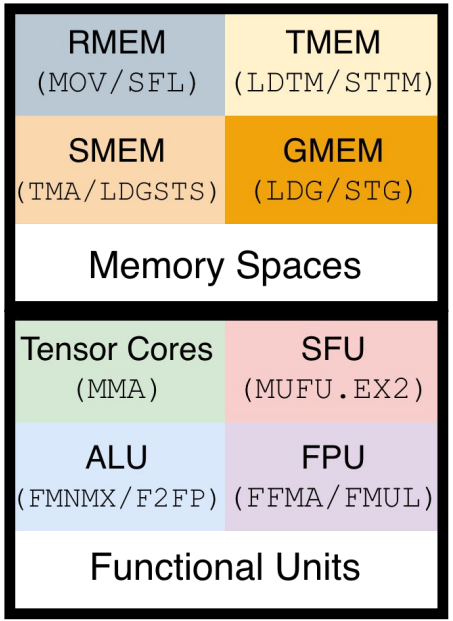


`LDTM`	:	Load TMEM ⟶ RMEM
`STTM`	:	Store RMEM ⟶ TMEM
`TMA`	:	Load/Store GMEM ⟷ SMEM `cp.async.bulk`
`LDGSTS`	:	Load GMEM ⟶ SMEM with `cp.async`
`LDG`	:	Load GMEM ⟶ RMEM
`STG`	:	Store RMEM ⟶ GMEM

`MMA`	:	Tensor Core Matrix Multiply Accumulate
`MUFU.EX2`	:	Exponentiation operation to the base 2
`FMNMX`	:	Floating-point min-max
`F2FP`	:	Floating-point conversion
`FADD`	:	Floating point add
`FMUL`	:	Floating point multiply
`FFMA`	:	Fused floating point multiply add

Attention Algorithm on the NVIDIA Blackwell SM

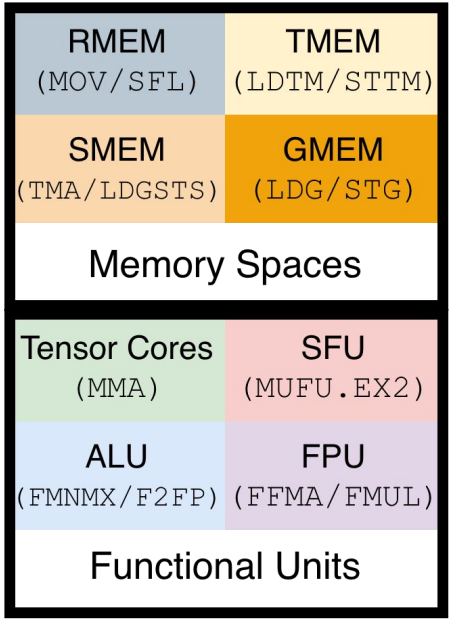
$$QK_{TMEM} = Q_{SMEM} @ K_{SMEM}^{T}$$



Attention Algorithm on the NVIDIA Blackwell SM

$$QK_{TMEM} = Q_{SMEM} \; @ \; K_{SMEM}^{T}$$

$$QK_{RMEM} = \mathtt{LDTM}(QK_{TMEM})$$

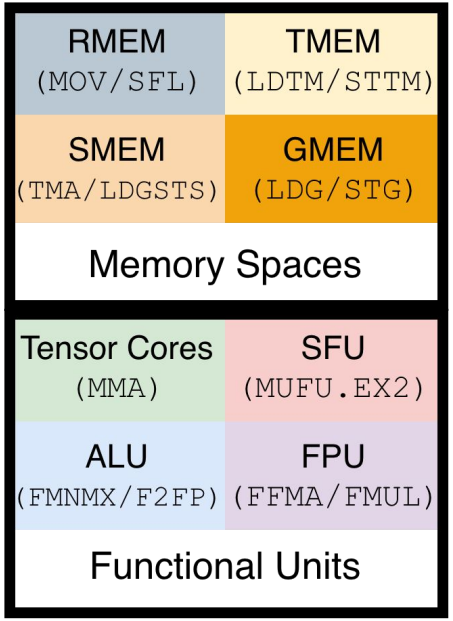


Attention Algorithm on the NVIDIA Blackwell SM

$$QK_{TMEM} = Q_{SMEM} \; @ \; K_{SMEM}^{T}$$

$$QK_{RMEM} = \mathtt{LDTM}(QK_{TMEM})$$

$$E_{RMEM} = \mathtt{Softmax}(QK_{RMEM})/\mathtt{sqrt}(H)$$



Attention Algorithm on the NVIDIA Blackwell SM

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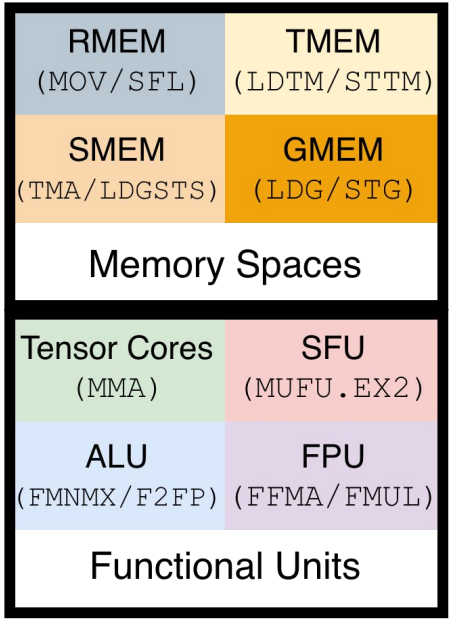
$$QK_{RMEM} = \mathrm{LDTM}(QK_{TMEM})$$

$$E_{RMEM} = \mathrm{Softmax}(QK_{RMEM})/\mathrm{sqrt}(H)$$

$$A_{RMEM} = \mathrm{Transform}(E_{RMEM})$$



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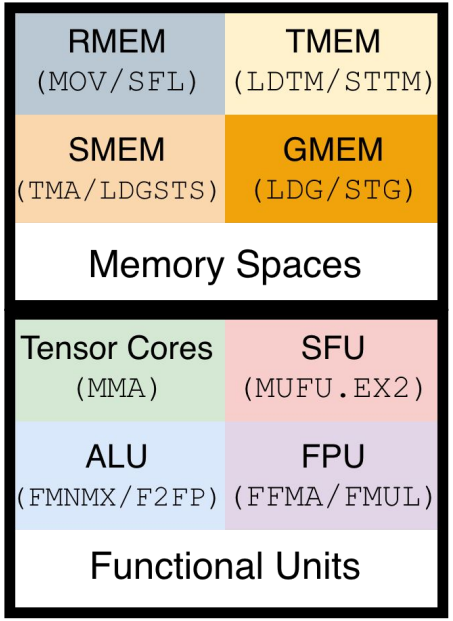
$$QK_{RMEM} = \texttt{LDTM}(QK_{TMEM})$$

$$E_{RMEM} = \texttt{Softmax}(QK_{RMEM})/\texttt{sqrt}(H)$$

$$A_{RMEM} = \texttt{Transform}(E_{RMEM})$$

$$A_{TMEM} = \texttt{STTM}(A_{RMEM})$$



Attention Algorithm on the NVIDIA Blackwell SM

$$QK_{TMEM} = Q_{SMEM} \; @ \; K_{SMEM}^{T}$$

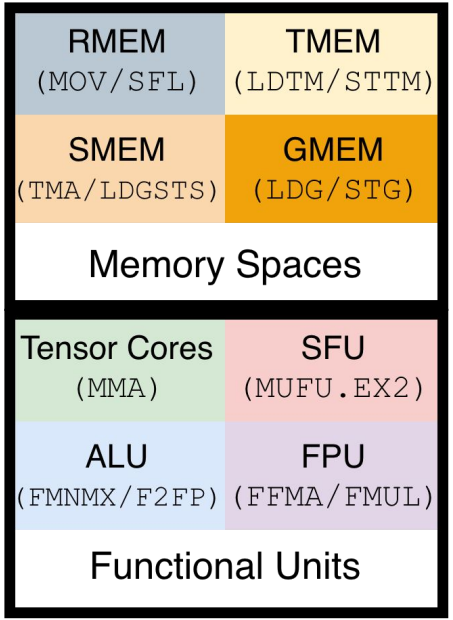
$$QK_{RMEM} = \texttt{LDTM}(QK_{TMEM})$$

$$E_{RMEM} = \texttt{Softmax}(QK_{RMEM})/\texttt{sqrt}(H)$$

$$A_{RMEM} = \texttt{Transform}(E_{RMEM})$$

$$A_{TMEM} = \texttt{STTM}(A_{RMEM})$$

$$AV_{TMEM} \mathrel{+}= A_{TMEM} \; @ \; V_{SMEM}$$



Attention Algorithm on the NVIDIA Blackwell SM

$$QK_{TMEM} = Q_{SMEM} @ K_{SMEM}^{T}$$

$$QK_{RMEM} = LDTM(QK_{TMEM})$$

$$E_{RMEM} = Softmax(QK_{RMEM})/sqrt(H)$$

$$A_{RMEM} = Transform(E_{RMEM})$$

$$A_{TMEM} = STTM(A_{RMEM})$$

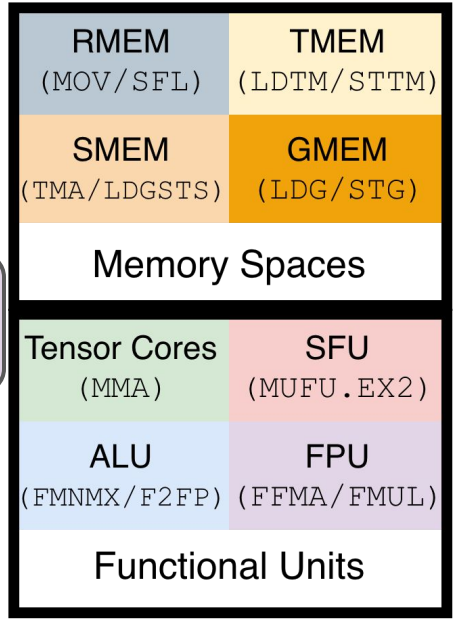
Correction

$$AV_{RMEM} = LDTM(AV_{TMEM})$$

$$AV_{RMEM} = AV_{RMEM} * RescaleFactor_{RMEM}$$

$$AV_{TMEM} = STTM(AV_{RMEM})$$

$$AV_{TMEM} \mathrel{+}= A_{TMEM} @ V_{SMEM}$$



Magic

Attention Algorithm on the NVIDIA Blackwell SM

$$QK_{TMEM} = Q_{SMEM} \; @ \; K_{SMEM}^{T}$$

$$QK_{RMEM} = \texttt{LDTM}(QK_{TMEM})$$

$$E_{RMEM} = \texttt{Softmax}(QK_{RMEM})/\texttt{sqrt}(H)$$

$$A_{RMEM} = \texttt{Transform}(E_{RMEM})$$

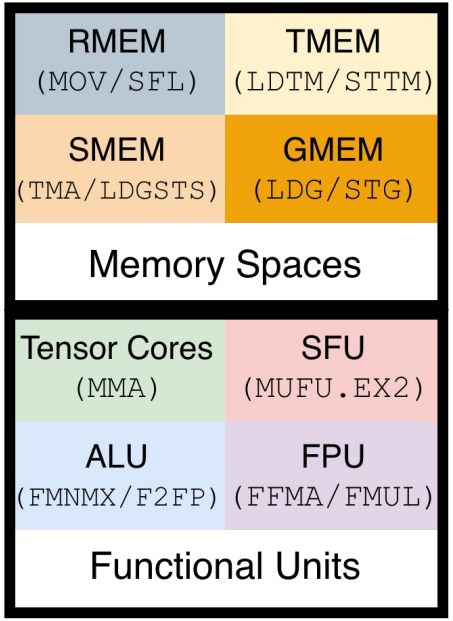
$$A_{TMEM} = \texttt{STTM}(A_{RMEM})$$

$$AV_{RMEM} = \texttt{LDTM}(AV_{TMEM})$$

$$AV_{RMEM} = \texttt{Correction}(AV_{RMEM})$$

$$AV_{TMEM} = \texttt{STTM}(AV_{RMEM})$$

$$AV_{TMEM} \mathrel{+}= A_{TMEM} \; @ \; V_{SMEM}$$



Agenda

- Attention Algorithm
- Tiling Attention on a GPU
- Convolution-inspired Runtime GQA Packing
- Attention on NVIDIA Blackwell Architecture
- Feeding the Datapath on NVIDIA Blackwell GB200

Feeding the Data Path on NVIDIA Blackwell

- **Within**-warp optimizations
  - Vectorization (e.g. `FMNMX3`, `FADD2`, `FFMA2`, `F2FP.*.BF16.F32.PACK_AB`)
  - Instruction level parallelism (ILP)
  - Software pipelining
- **Across**-warp asynchrony (Warp-specialization)
  - 1xSoftmax
  - 2xSoftmax
  - Exp-Transform
  - 2xQ
- **Efficient** use of Tensor Memory (TMEM)
  - `256KB` of TMEM per SM
  - Many different partitioning possible with different trade-offs

Feeding the Data Path on NVIDIA Blackwell

- **Within**-warp optimizations
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- **Efficient** use of Tensor Memory (TMEM)
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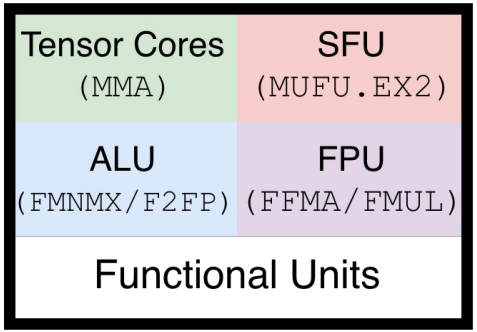
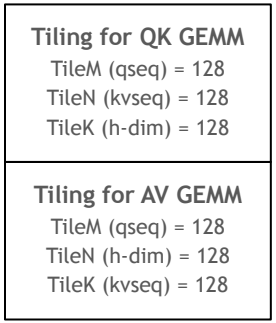
Roofline Analysis - NVIDIA GB200

Machine (GB200)	Peak rate (per-GPU)	Peak rate (per-SM, per-Cycle)
MMA **(BF16)**	2.5 PFLOPS	8192 FLOPS/SM/Cycle
MUFU.EX2	5 TEXPOPS	16 EXOPS/SM/Cycle
FFMA	80 TFLOPS	256 FLOPS/SM/Cycle
FMUL	40 TFLOPS	128 FLOPS/SM/Cycle

Attention Algorithm	Total OPS (per Tile)	Cycles (Total OPS/Peak rate)
MMA QK	2*128*128*128	**512**
MMA AV	2*128*128*128	**512**
MUFU.EX2	128*128	**1024**
FFMA ($x*scale - x_{max}$)	2*128*128	**128**
FMUL ($AV_{corr} = AV * rescale$)	128*128	**128**



NVIDIA Blackwell (**GB200**)



Attention Tiling Details

Within-Warp Optimizations - `Correction`

We can achieve nearly perfect Instruction-Level Parallelism (ILP) for `Correction` because of:

- Straight-forward dependency chains
- Double-buffering in RMEM, prefetching one stage from TMEM, and interleaving `LDTM`, `FMUL`, and `STTM`
- Exposes only a part of `LDTM` at the start and `STTM` at the end

Attention Algorithm	Total OPS (per Tile)	Cycles (Total OPS/Peak rate)
MMA QK	2*128*128*128	512
MMA AV	2*128*128*128	512
MUFU.EX2	128*128	1024
FFMA ($x*scale - x_{max}$)	2*128*128	128
FMUL (AV_{corr} = AV * rescale)	128*128	**128**


```
LDTM.x16 R20, tmem[UR5+0x100] ;
LDTM.x16 R4, tmem[UR5+0x110] ;
FMUL2 R20, R20.F32x2.HI_LO, R53.reuse.F32 ;
...
FMUL2 R6, R6.F32x2.HI_LO, R53.reuse.F32 ;
STTM.x16 tmem[UR5+0x100], R20 ;
FMUL2 R8, R8.F32x2.HI_LO, R53.reuse.F32 ;
LDTM.x16 R36, tmem[UR5+0x120] ;
FMUL2 R10, R10.F32x2.HI_LO, R53.F32 ;
...
FMUL2 R18, R18.F32x2.HI_LO, R53.F32 ;
STTM.x16 tmem[UR5+0x110], R4 ;
LDTM.x16 R20, tmem[UR5+0x130] ;
FMUL2 R36, R36.F32x2.HI_LO, R53.reuse.F32 ;
...
FMUL2 R22, R22.F32x2.HI_LO, R53.reuse.F32 ;
STTM.x16 tmem[UR5+0x120], R36 ;
FMUL2 R24, R24.F32x2.HI_LO, R53.reuse.F32 ;
LDTM.x16 R4, tmem[UR5+0x140] ;
FMUL2 R26, R26.F32x2.HI_LO, R53.F32 ;
...
FMUL2 R34, R34.F32x2.HI_LO, R53.F32 ;
STTM.x16 tmem[UR5+0x130], R20 ;
LDTM.x16 R36, tmem[UR5+0x150] ;
FMUL2 R4, R4.F32x2.HI_LO, R53.reuse.F32 ;
...
FMUL2 R38, R38.F32x2.HI_LO, R53.reuse.F32 ;
STTM.x16 tmem[UR5+0x140], R4 ;
FMUL2 R40, R40.F32x2.HI_LO, R53.reuse.F32 ;
LDTM.x16 R20, tmem[UR5+0x160] ;
FMUL2 R42, R42.F32x2.HI_LO, R53.F32 ;
...
FMUL2 R50, R50.F32x2.HI_LO, R53.F32 ;
STTM.x16 tmem[UR5+0x150], R36 ;
LDTM.x16 R4, tmem[UR5+0x170] ;
FMUL2 R20, R20.F32x2.HI_LO, R53.reuse.F32 ;
...
FMUL2 R6, R6.F32x2.HI_LO, R53.reuse.F32 ;
STTM.x16 tmem[UR5+0x160], R20 ;
FMUL2 R8, R8.F32x2.HI_LO, R53.F32 ;
...
FMUL2 R18, R18.F32x2.HI_LO, R53.F32 ;
STTM.x16 tmem[UR5+0x170], R4 ;
```

Our GPUs 💚 beautiful SASS

Within-Warp Optimizations - Softmax

Achieving a good ILP for the Softmax is difficult because:

- Dependencies on LDTM and FMNMX, before the first MUFU
- Too many instruction kinds in the mix
- Too many MUFU and too few ALU and FPU OPS

Attention Algorithm	Total OPS (per Tile)	Cycles (Total OPS/Peak rate)
MMA QK	2*128*128*128	512
MMA AV	2*128*128*128	512
MUFU.EX2	128*128	**1024**
FFMA (x*scale - x_{max})	2*128*128	**128**
FMUL (AV_{corr} = AV * rescale)	128*128	128

MUFU block theoretical

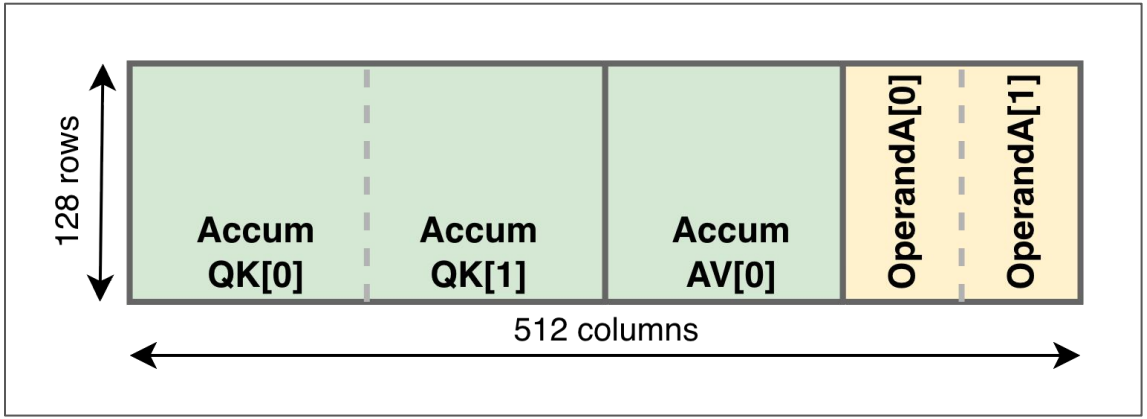
MUFU block in Attention Softmax

MUFU block in Attention Softmax with partial emulation

Feeding the Data Path on NVIDIA Blackwell

- **Within**-warp optimizations
  - Vectorization (e.g. `FMNMX3`, `FADD2`, `FFMA2`, `F2FP.*.BF16.F32.PACK_AB`)
  - Instruction level parallelism (ILP)
  - Software pipelining
- **Across**-warp asynchrony (Warp-specialization)
  - 1xSoftmax
  - 2xSoftmax
  - Exp-Transform
  - 2xQ
- **Efficient** use of Tensor Memory (TMEM)
  - `256KB` of TMEM per SM
  - Many different partitioning possible with different trade-offs

Tensor Memory on Blackwell

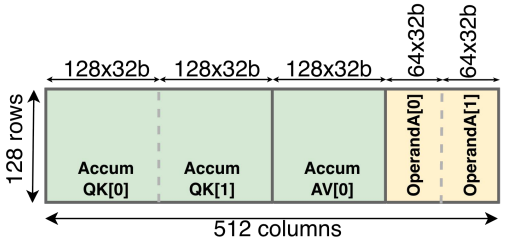


- **Easier pipelining:** promotes creating “crazier” warp-specialized schedules
- **Faster data transfer:** allows warpgroups to communicate data
- **Lower register pressure:** Offloads the accumulators from RMEM to TMEM

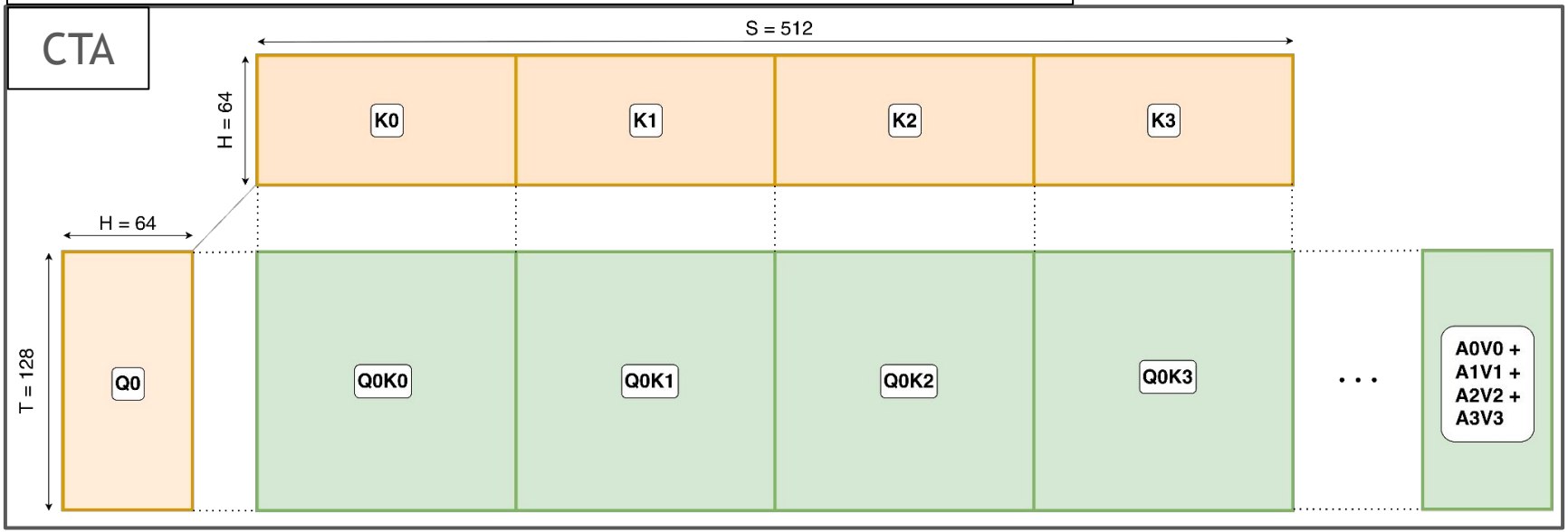
1xSoftmax, 2xSoftmax, Exp-Transform

1SM loads 1 Q tile, loads 4 KV tiles

- 1xQ tile per SM
- Lower SMEM usage for Q tiles, Higher SMEM usage for KV
  - Latency hiding by PreFetching more KV tiles
- TMEM partitioning
  - Two slots for AccumQK,
  - Two slots for OperandA
  - One slot for AccumAV

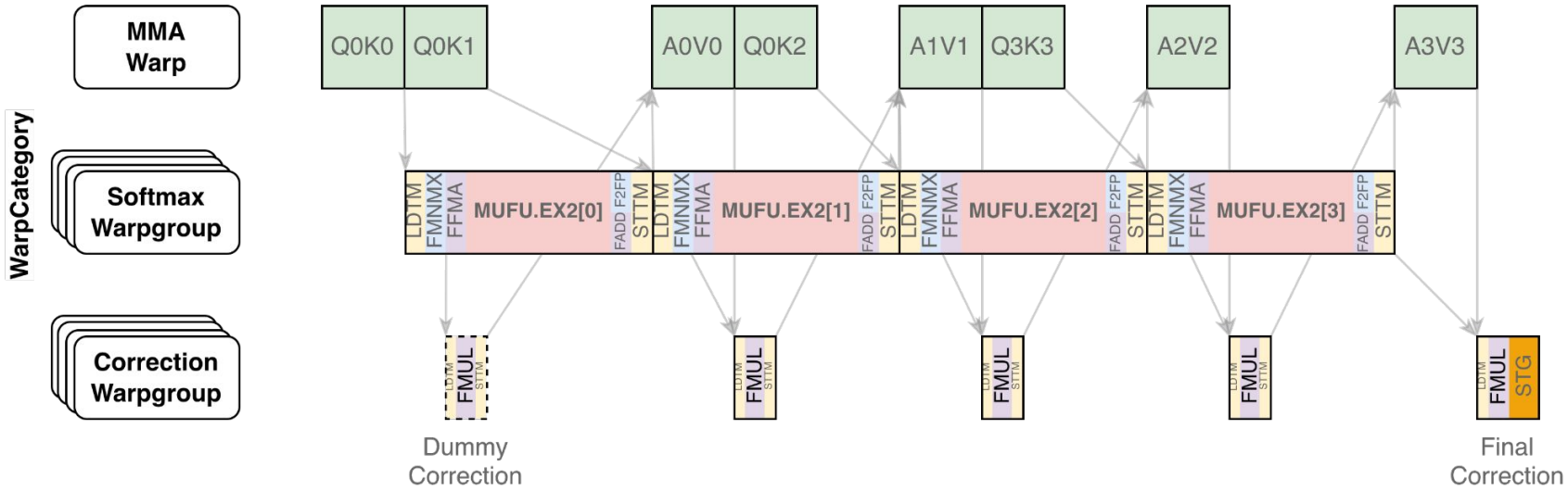


TMEM Partitioning



1xSoftmax

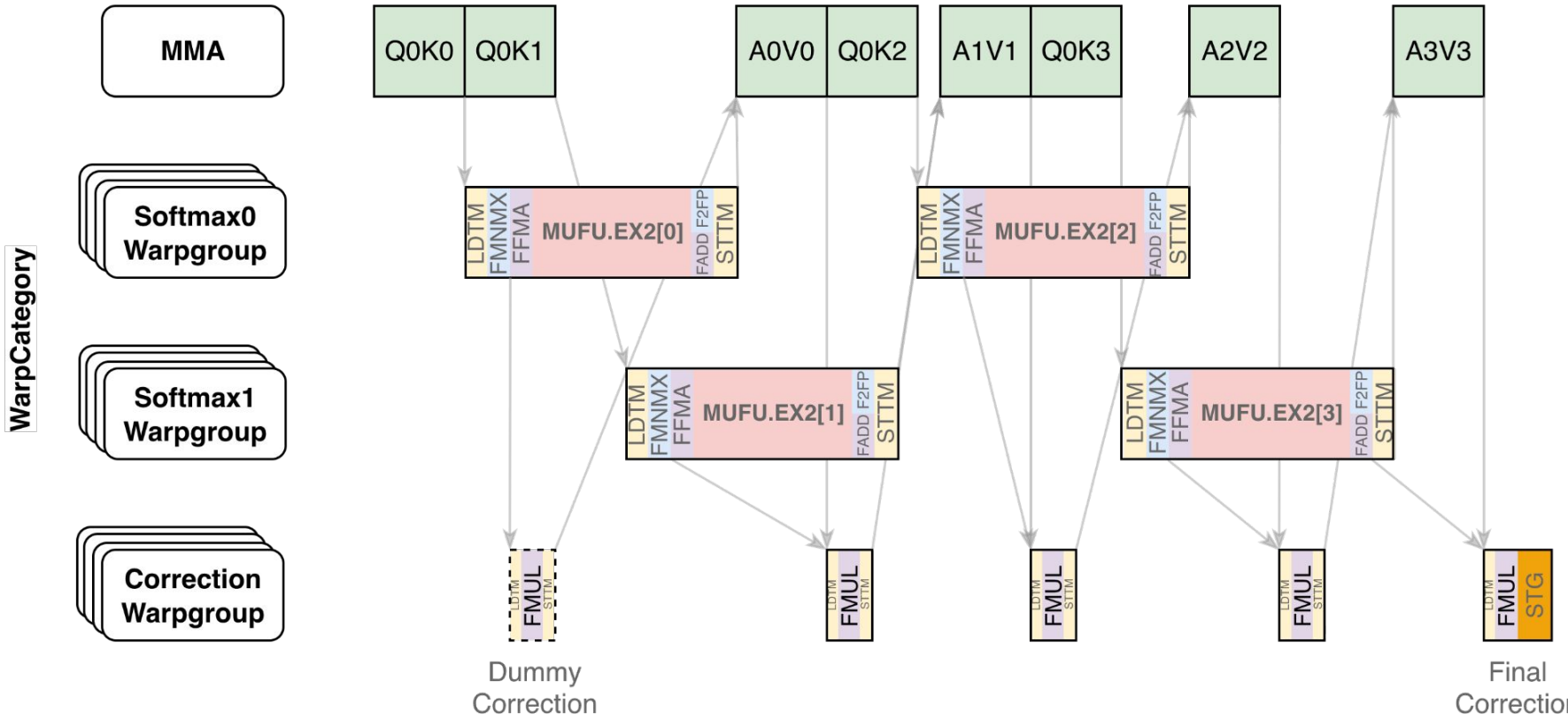
1SM loads 1 Q tile, loads 4 KV tiles, computes 4 softmax tiles with 1 softmax warpgroup



- Simplest schedule using all of the Blackwell features with minimal pipelining/asynchrony
- Excellent initial schedule for Blackwell attention optimization
- Effective for certain attention shapes
- Inefficient use of `MMA` and `MUFU` units

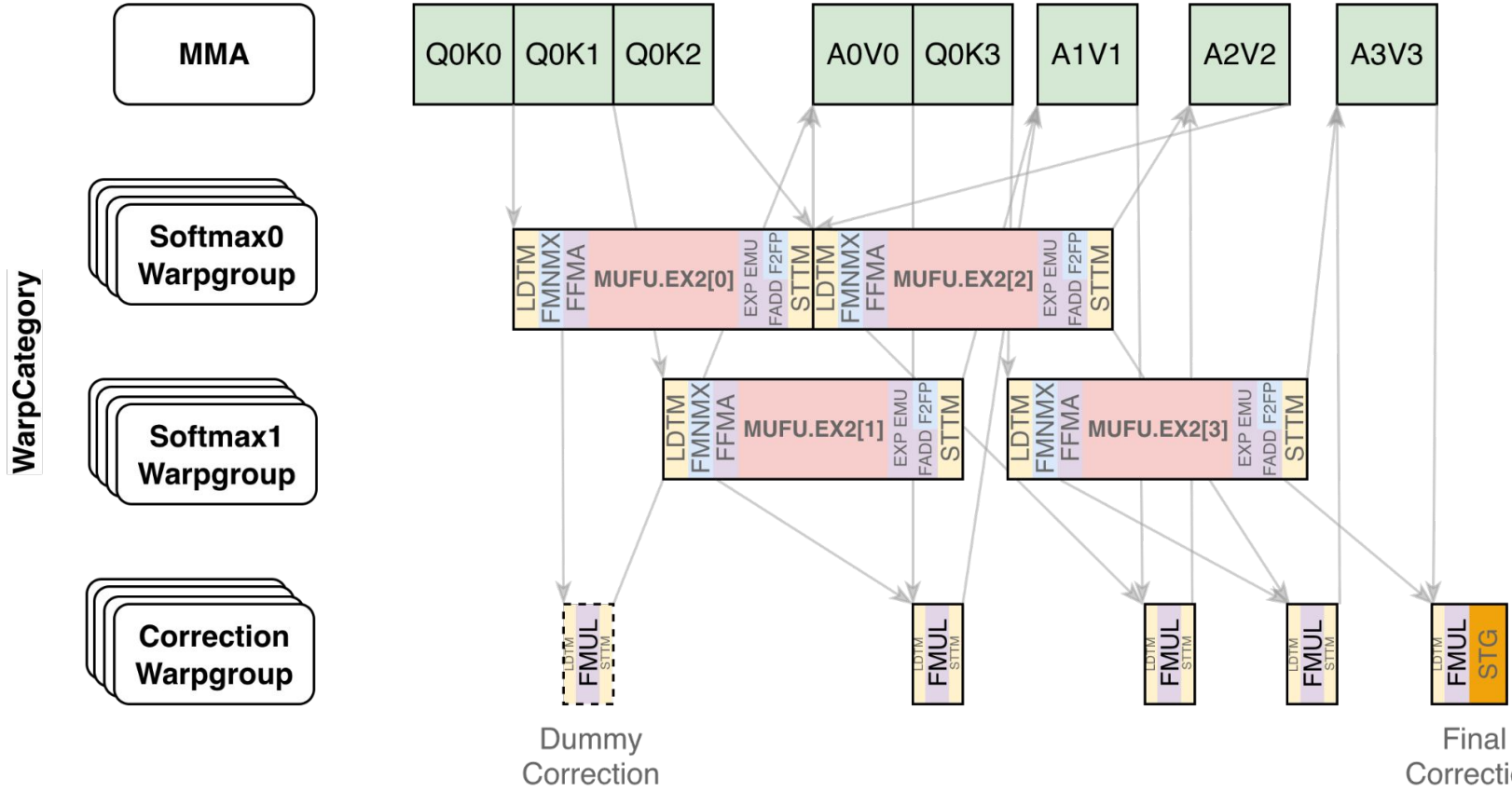
2xSoftmax

1SM loads 1 Q tile, loads 4 KV tiles, computes 4 softmax tiles with 2 softmax warpgroup



2xSoftmax - PreCompute QKs and EXP Emulation

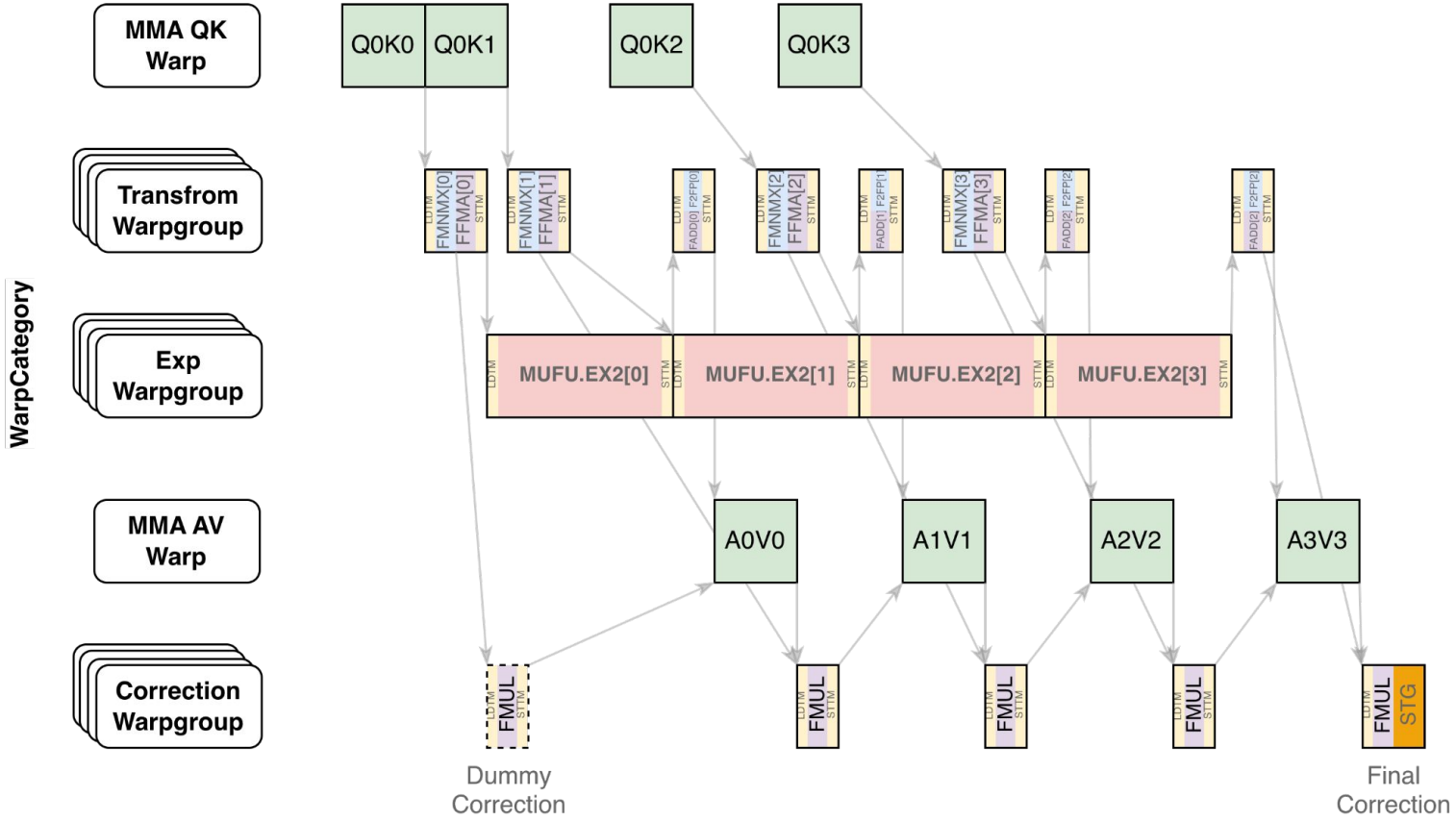
1SM loads 1 Q tile, loads 4 KV tiles, computes 4 softmax tiles with 2 softmax warpgroup



Exp-Transform

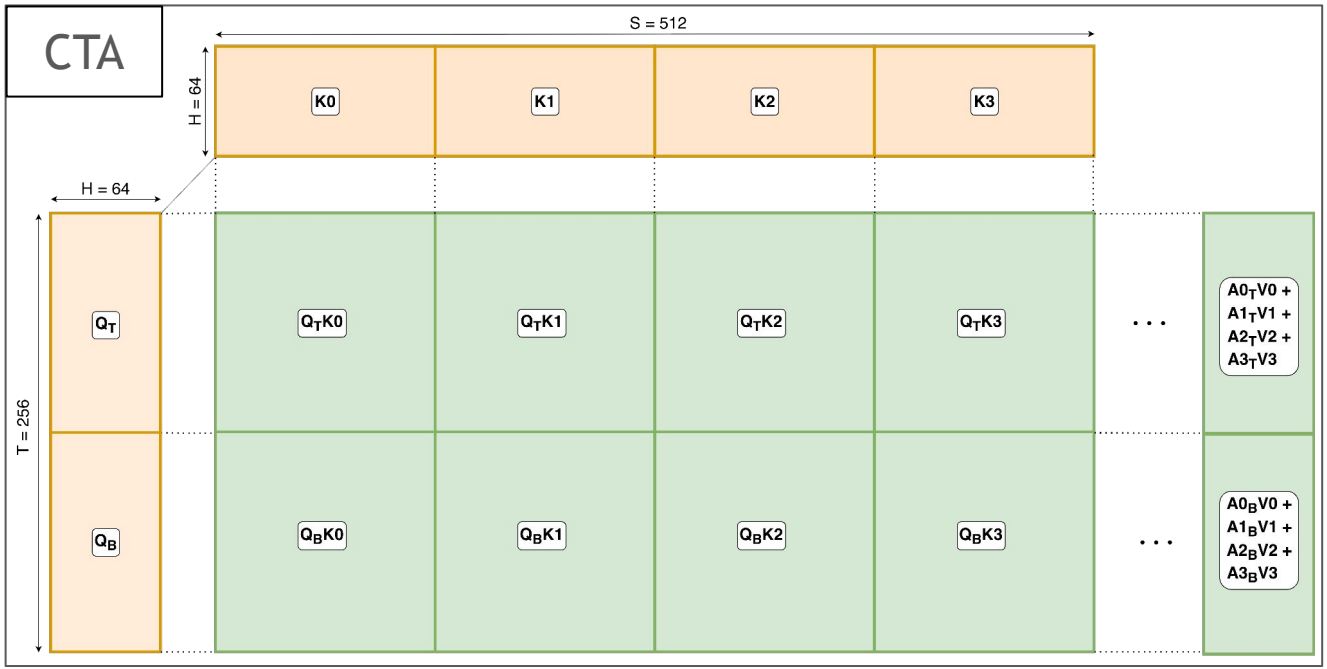
Make every functional unit feel special

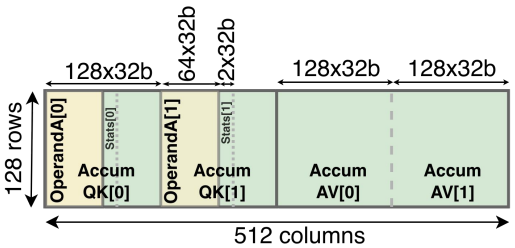
1SM loads 1 Q tile, iterates 4 KV tiles, computes 4 softmax tiles with a Transform and an Exp warpgroup



2xQ

1SM loads 2 Q tile, loads 4 KV tiles, computes 8 softmax tiles with 2 softmax warpgroups



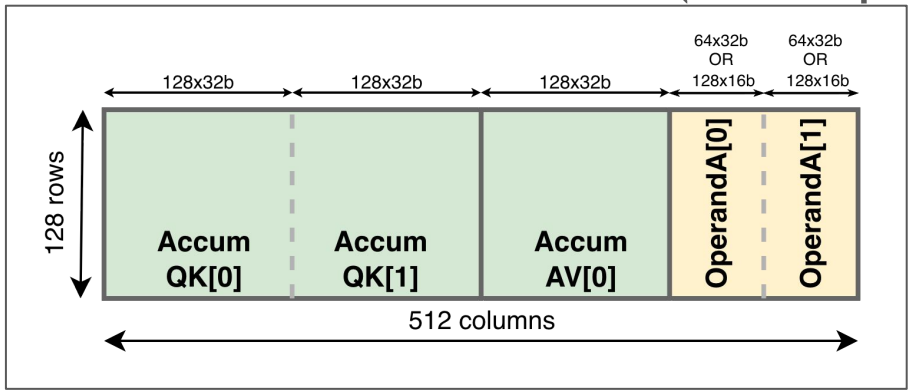


TMEM Partioniting

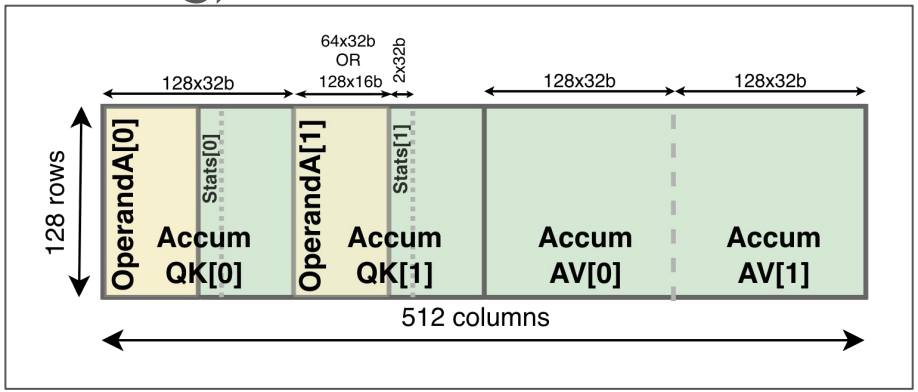
Warp-specialized Kernel Schedule first pioneered in **CUTLASS example 77, Markus Hoehnerbach** et. al. and now optimized in **FA4, Ted Zadouri** et. al.

Tensor Memory on Blackwell

(TMEM partitioning)



Non Overlapping

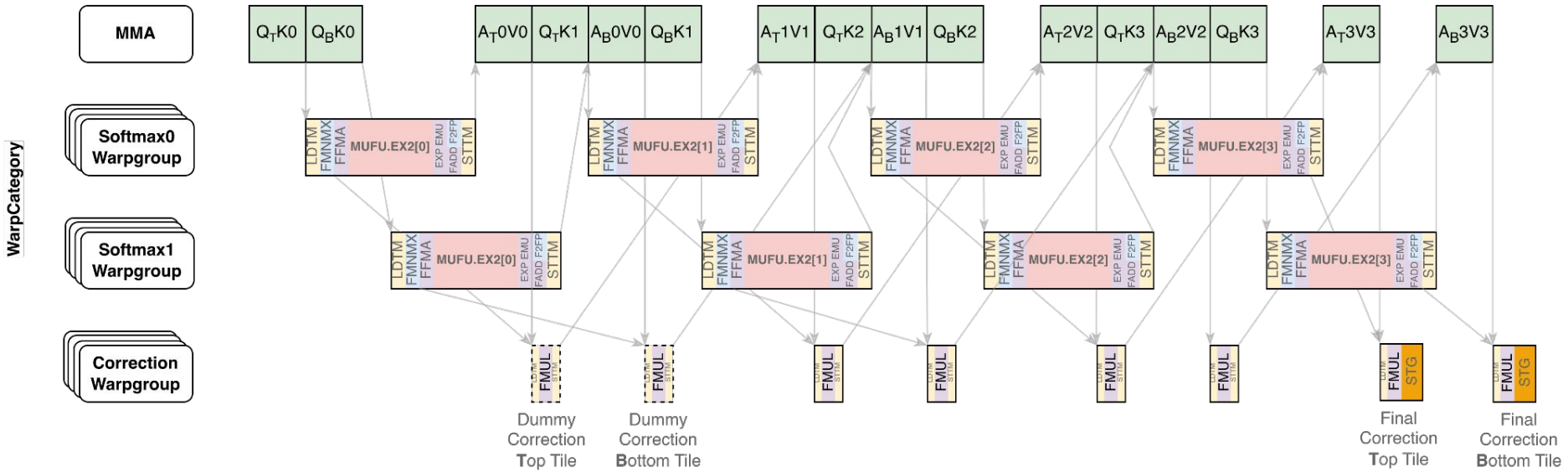


Overlapping

Non Overlapping TMEM Sections	Overlapping TMEM Sections
Dedicated TMEM slots for 2xAccumQK and 2xOperandA **Dedicated** TMEM slots for 1xAccumAV	**Overlapping** TMEM slots for 2xAccumQK and 2xOperandA **Dedicated** TMEM slots for 2xAccumAV
One slot for **AccumAV**	**Two** slots for **AccumAV**
AccumQK[i] stay in **critical region** for **shorter** duration	**Accum QK[i]** stay in **critical region** for **longer** duration

2xQ

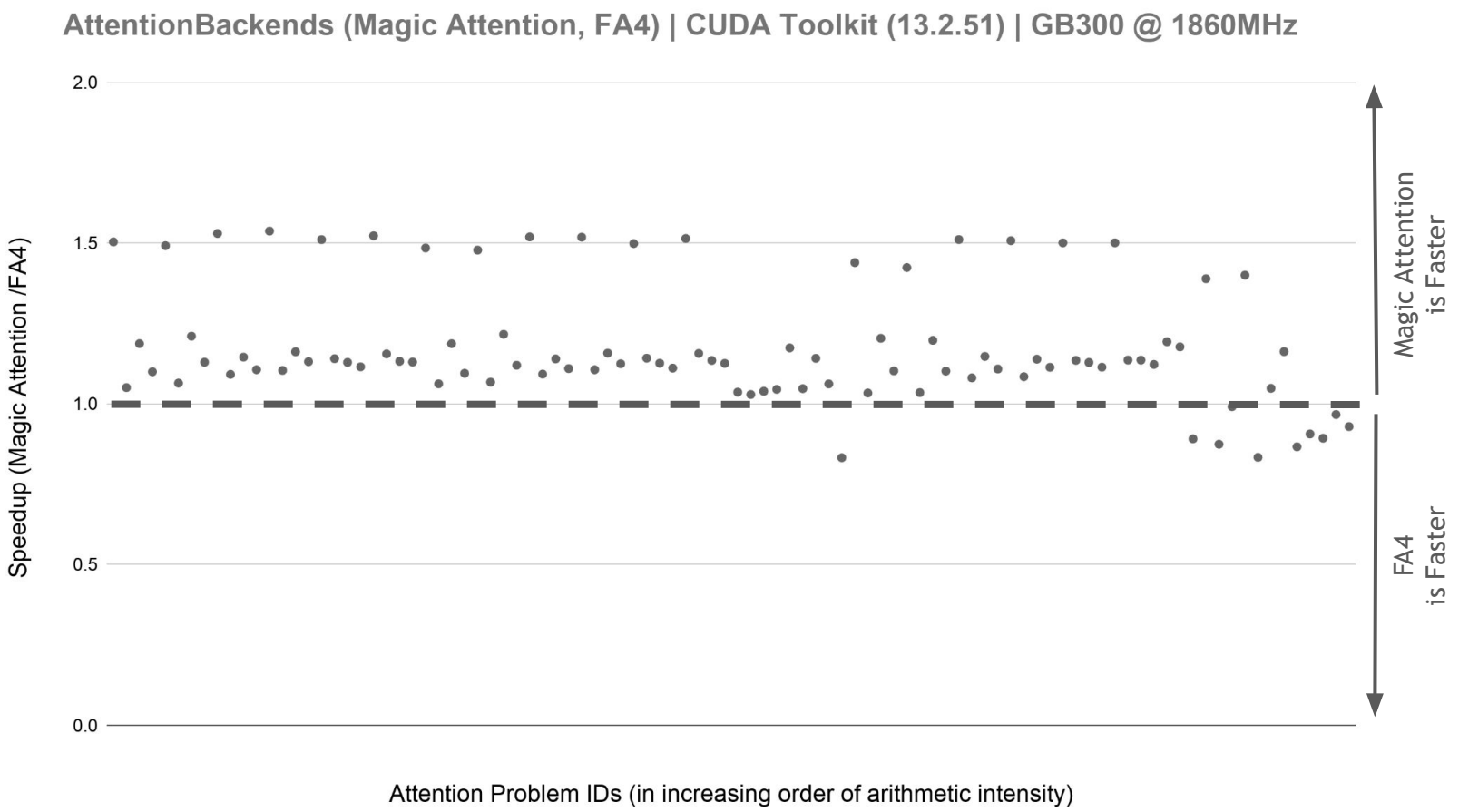
1SM loads 2 Q tile, loads 4 KV tiles, computes 8 softmax tiles with 2 softmax warpgroups



Warp-specialized Kernel Schedule first pioneered in **CUTLASS example 77, Markus Hoehnerbach** et. al. and now optimized in **FA4, Ted Zadouri** et. al.

Magic

GB300


AttentionBackends (Magic Attention, FA4) | CUDA Toolkit (13.2.51) | GB300 @ 1860MHz
Speedup (Magic Attention /FA4)
2.0
1.5
1.0
0.5
0.0
Magic Attention is Faster
FA4 is Faster
Attention Problem IDs (in increasing order of arithmetic intensity)

Magic

Thank you!

Magic